Conductor-gap replication: results and limits

Date: 2026-10-25. Follow-up to PR #2430. Classification: bounded controlled measurement, with exact arithmetic and accounting checks. Every scientific verdict below uses the frozen design; wall time is descriptive L0 only.

Assessment. This is stronger bounded evidence for similar counted-rho cost across the gap. All five levels pass the nominal +/-5% rule; four pass the simultaneous correction, while the crater remains inconclusive. The CM construction comparisons change setup work and preserve the non-base solver behavior. A general endomorphism-dependent hardness conclusion is still unestablished.

Balanced rho panel

Depth	S2/S1	90% whole-class interval	98% interval	Nominal +/-5% rule	Simultaneous rule
0	1.01113	[0.98304, 1.04267]	[0.97429, 1.05660]	passes	inconclusive
1	0.99870	[0.98353, 1.01458]	[0.97776, 1.02110]	passes	passes
2	1.00213	[0.98006, 1.02301]	[0.97069, 1.03051]	passes	passes
3	0.98590	[0.96888, 1.00149]	[0.96095, 1.00689]	passes	passes
4	0.99944	[0.97666, 1.02920]	[0.97062, 1.04485]	passes	passes

Derived observations. Each ratio first averages three S2 volcano means and divides by S1 within one class, then takes the geometric mean across eight classes. Each level has 192 separately executed one-target solves per volcano. All 30,720 primary observations have equal effort by level; the 8,192 control observations and 2,944 BSGS checks are separate.

Effort is balanced, while target indices are deliberately shared across transported vertices: 192/64/32/16/8 indices per vertex at depths 0..4. The floor therefore has eight distinct scalar indices across its 24 vertices, with separate seeded solver executions. The whole-class bootstrap retains this dependence; rows are not treated as IID targets.

The secondary equal-level ratio is 0.99943, 95% interval [0.98606, 1.01455]. It does not substitute for the five primary level results. All class means, individual S2/S1 ratios, 90/95/98/99% intervals and exact raw-file hashes are in panel-A.json.

Class-level observations vary: class 7 has S2/S1 ratios 1.11983 at the crater and 1.11897 at depth four. The pooled result does not establish uniform equivalence for each class; these observed deviations remain visible in the raw class means and are not explained away by pooling.

Controls and validation

Check	Native observation
Real positive control, plain rho / negation rho	1.36704, 95% [1.34569, 1.39371]
A/A, same method on matching workloads	0.99990, 95% [0.96553, 1.03431]; 1280 matched pairs
Coordinate model 1 / model 0	0.99988, 95% [0.97465, 1.02320]
All control gates	true
BSGS costs equal across transported instances per target	true
Planned A / B records retained	41856 / 1664
Independent scalar checks	43520 correct; 43520 deliberately changed scalars rejected
Independent input and graph checks	12 classes; 1472 vertices; 1440 Phi2 edges; 768 floor quadratic factors irreducible

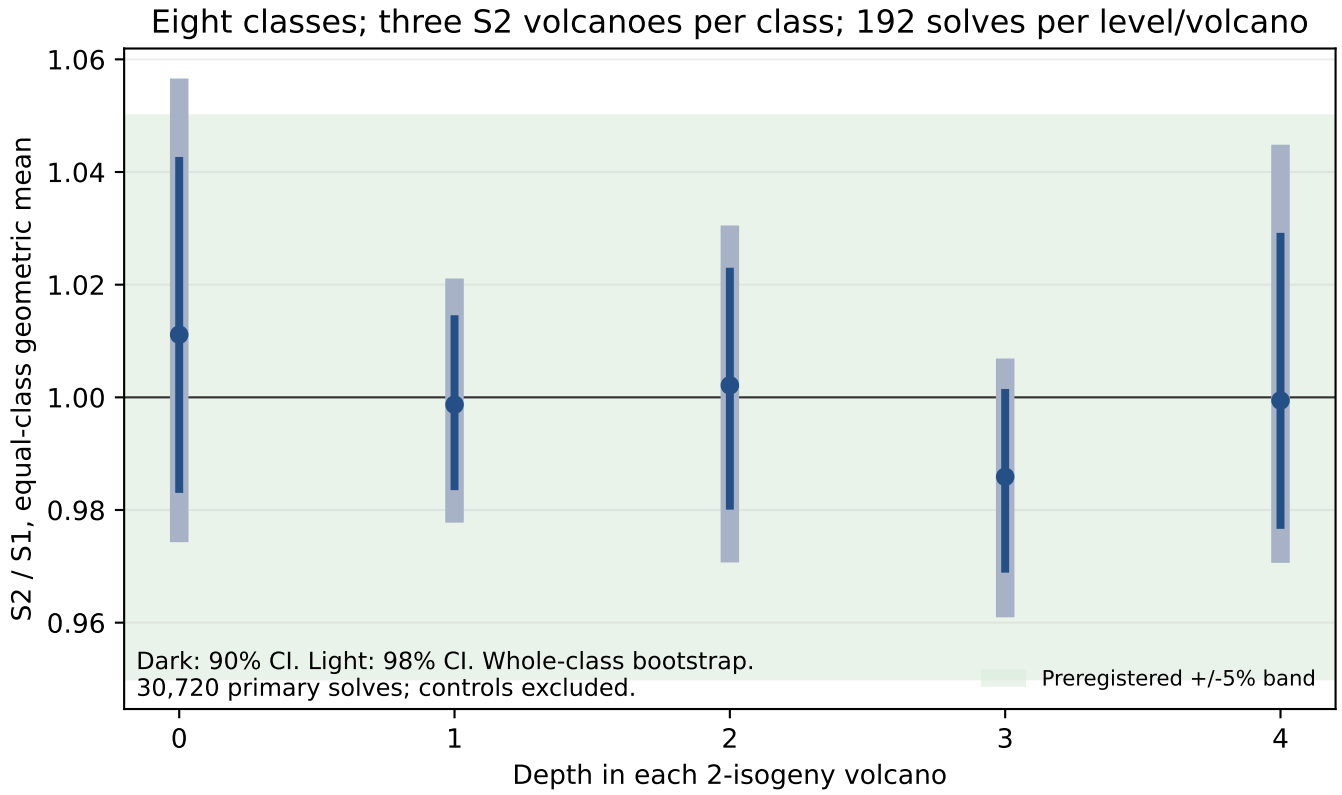


Figure 1: Recorded ratios; light bars are 98% and dark bars 90% whole-class bootstrap intervals.

The measurement binary independently checked every recovered scalar; exact ecbench replay re-executed every deterministic observation. A separately written C++ implementation checks curve and subgroup primality, point membership, subgroup orders, recovered scalars, and rejection of scalar+1. It also recomputes the headline ratios from the exported raw-record CSV, using no library arithmetic or native analyzer code. Its bootstrap uses independent mt19937_64 draws, so it is a Monte Carlo cross-check rather than byte-identical reproduction of Rust StdRng intervals.

Depth	Native geometric mean	Independent recompute	Independent 90% interval
0	1.01113	1.01113	[0.98296, 1.04312]
1	0.99870	0.99870	[0.98384, 1.01444]
2	1.00213	1.00213	[0.97950, 1.02343]
3	0.98590	0.98590	[0.96839, 1.00175]
4	0.99944	0.99944	[0.97666, 1.03054]

These checks use a separate implementation on the same host. They do not claim a second physical runner, isolated timing, a fully priced IC/rho win, or cryptographic-size validation. The committed audit workflow can repeat the archive replay on another host. Machine-readable certificates: scalar and summary checks, geometry, export sources.

CM-aware collector panel

Panel B uses four additional prime-order classes, two each for CM fields -19 and -43, three sampled floor curves per class and an isomorphic model of every curve. The large-conductor kernels have abscissa field degree at least (N-1)/8. This is separate from the twist-rational construction used for the deep volcanoes.

On every model, scalar and generic-recurrence orbit construction feed the same relation collector and matched rho.negation reference. Direct degree-5 or degree-11 CM maps are additionally tested on maximal-order craters. These small self-maps act within a crater; transport across the large-N conductor gap remains a separate cost and construction problem. All 32 prepared models have exact scalar/recurrence base equality; direct map equality also holds on all eight

crater models. The recurrence P[k+2] = P[k+1] - [q]P[k], after one scalar initialization, is available on both sides. A faster base constructor does not by itself establish a faster unknown-scalar algorithm or a conductor-gap advantage.

The factoring certificate IDC1ha50d552f2f9b8cb1 is retained in recurrence-factorization.json and checked with 128 additional checker-seeded points. It evaluates z(lambdaˆ2-lambda+q) = lambda(lambda z)-lambda z+qz over the certificate field. The group recurrence additionally requires the per-instance characteristic-polynomial relation and prime-order point checks; a finite polynomial evaluation is not a complete CM geometry proof. The prior study's conditional Lean theorem recurrence_matches_orbit gives the two-state invariant; the present native preflight checks its actual point realization on all prepared models.

Factor-base construction	Recorded attempts	Mean charged group GAE
ic-recurrence	512	768.00000
ic-scalar	512	5208.00000
ic-velu	128	90.00000

These are construction-phase group-work counts. Direct-map counts cover craters only and omit unpriced field work; they are not zero-cost or fully priced speed claims. Recorded cold-cost floors and all matched cells remain in the native summaries below.

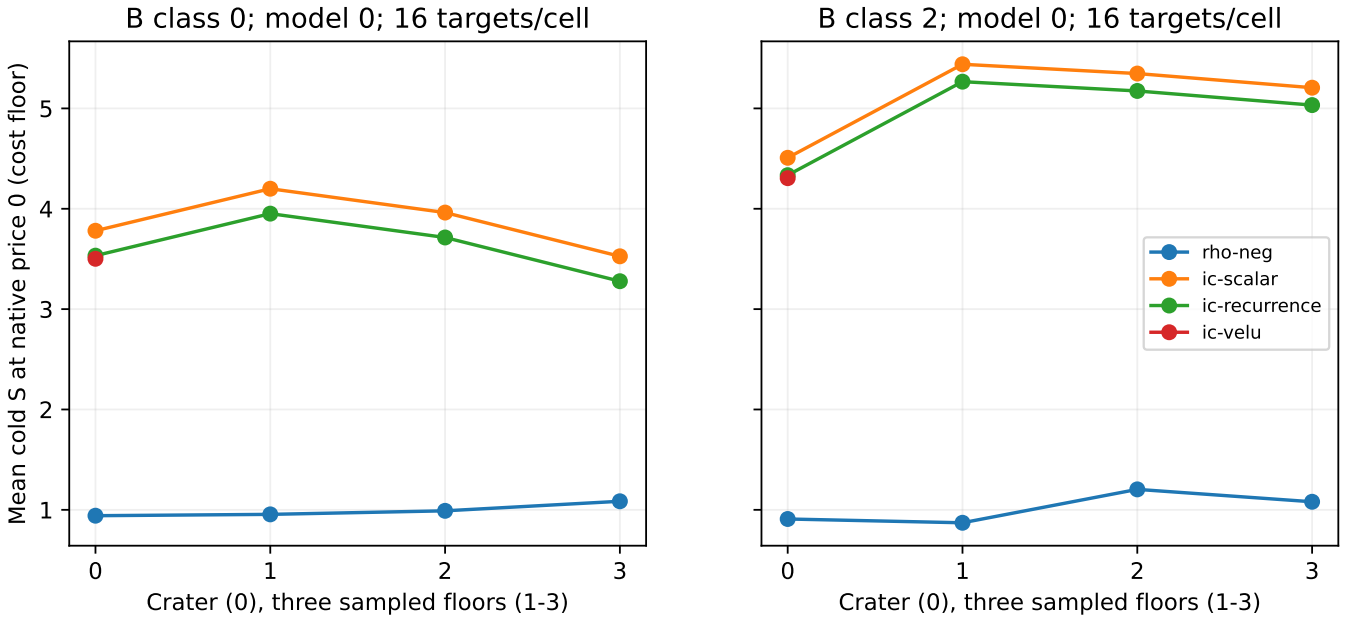


Figure 2: Recorded cost floors for model 0 in one class of each CM field. All cells and pricing bands are retained in native JSON.

All 1,664 B attempts, including the matched references, retain their statuses. panel-B.json gives every class, volcano, coordinate model and arm; mode invariants checks matching seeds, recovered scalars, queries, relations, ranks and every non-base counted phase between constructions. No censored workload is replaced by a success-only average.

Every arm, including rho, is evaluated at 0, 0.1 and 1 GAE per unpriced native operation using ecbench's common-band implementation. All per-curve bands, missing counts, uncharged native counters, phase costs, query/rank totals, peak memory and statuses are in diagnostics-A.json and diagnostics-B.json. Unknown prices stay unknown. A quotient of two cost floors is a diagnostic, not a bound on the full-cost quotient. No fastest-tool or full-cost method ranking is admitted here; no Magma/msolve comparison has been performed.

The band prices the reported counters. It cannot supply missing counts: the CM action's scalar modular arithmetic, parts of twist-kernel square-root search and codomain normalization do not have complete separate operation counters. This setup-accounting gap is distinct from an unknown conversion price. Neither it nor unmeasured native work is assigned zero physical cost.

Input coverage and scope

A class	CM discriminant	Field prime p	Prime subgroup r	Conductor gap N
0	-11	3657893441	19051481	2273
1	-11	16581927113	460610153	4799
2	-19	3957186017	61830919	1801
3	-19	17039714201	608560159	3719
4	-43	4099664089	60289321	1217
5	-43	16983164089	4245795977	2477
6	-67	4262310593	46329463	997
7	-67	17175115961	1073443253	1993

B class	CM discriminant	Field prime p	Prime subgroup r	Gap N	Kernel-abscissa field degree
0	-19	320092037	320091991	8209	1368
1	-19	320917397	320916971	8219	4109
2	-43	728661677	728661587	8233	1029
3	-43	729368027	729367999	8237	4118

All exact ICV1, EC1 and generator identities are in the registered A identities and B identities, and every frozen plan reports registered models. Independent geometry checks verify the field/order discriminant identity, prime gaps, exact kernel-field degrees, disjoint volcano j sets, all rational 2-isogeny neighbors and floor completeness. Scalar-order witnesses exceeding the Hasse-interval width certify the claimed full order. Endomorphism-order interpretation still uses standard ordinary-CM/order-lattice mathematics; it is not a new full Lean proof.

The diagrams use the repository's standard volcano renderer. Labels map to full identities in the adjacent .identities.json files.

Interpretation limits. Eight deterministically selected A classes are a fixed cohort, not a representative random sample of ordinary curves. The whole-class bootstrap quantifies sensitivity within that cohort; its nominal coverage is not a theorem about all CM fields. A is at 32/34 field bits and depth four, explicitly smaller than the parent's depth-six, 52-bit class. B has no deep 2-volcano and independently chosen floor generators; it does not supply hard-gap target transport. Cofactor-one CM collection is not implemented on the deep panel. Prime-field F4/SAT descent, a general exploitable endomorphism algorithm, complete common pricing, larger independent families and cryptographic-scale coverage remain open. No scaling exponent, hardness theorem, impossibility claim or deployed-curve security conclusion follows, and a possible intermediate algebraic advantage is not ruled out by a generic-rho result.

Preflight corrections, provenance and reproduction

The first B fixture set failed direct-map preflight because its twist lacked the required small CM kernel; it also contained an anomalous trace-one case. It was never used for solver measurements. The rejected v1 fixtures and failure remain intact. The dated amendments add map eligibility and exclude anomalous curves before primary data. All 128 disjoint calibration solves verify and replay, and remain excluded from inference. Calibration estimated about 94 minutes including startup and replay; the shared main guard was raised to 120 minutes before primary observations. No sample count or inference threshold changed after data. The actual main command span was about 84.3 minutes; the conservative calibration overestimated replay overhead. Timing here describes execution and resource use, not an admitted algorithm-performance comparison.

The original protocol, dated amendments, committed source, exact 64 primary specs, native plans, compiler/binary hashes and final preflight are retained. The seven-command main card, its receipt, verbatim executor relay and director assessment record execution separately from scientific interpretation. The calibration has the same artifacts. The assessment documents the canonical task-card checker's post-execution lint/output-order limitations without altering its code or evidence.

A post-measurement catalogue check caught integer rounding in the initial JavaScript registry merge. The lossless Rust repair restores 362 legacy models from the exact prior snapshot and preserves all 2,976 experiment models. The original

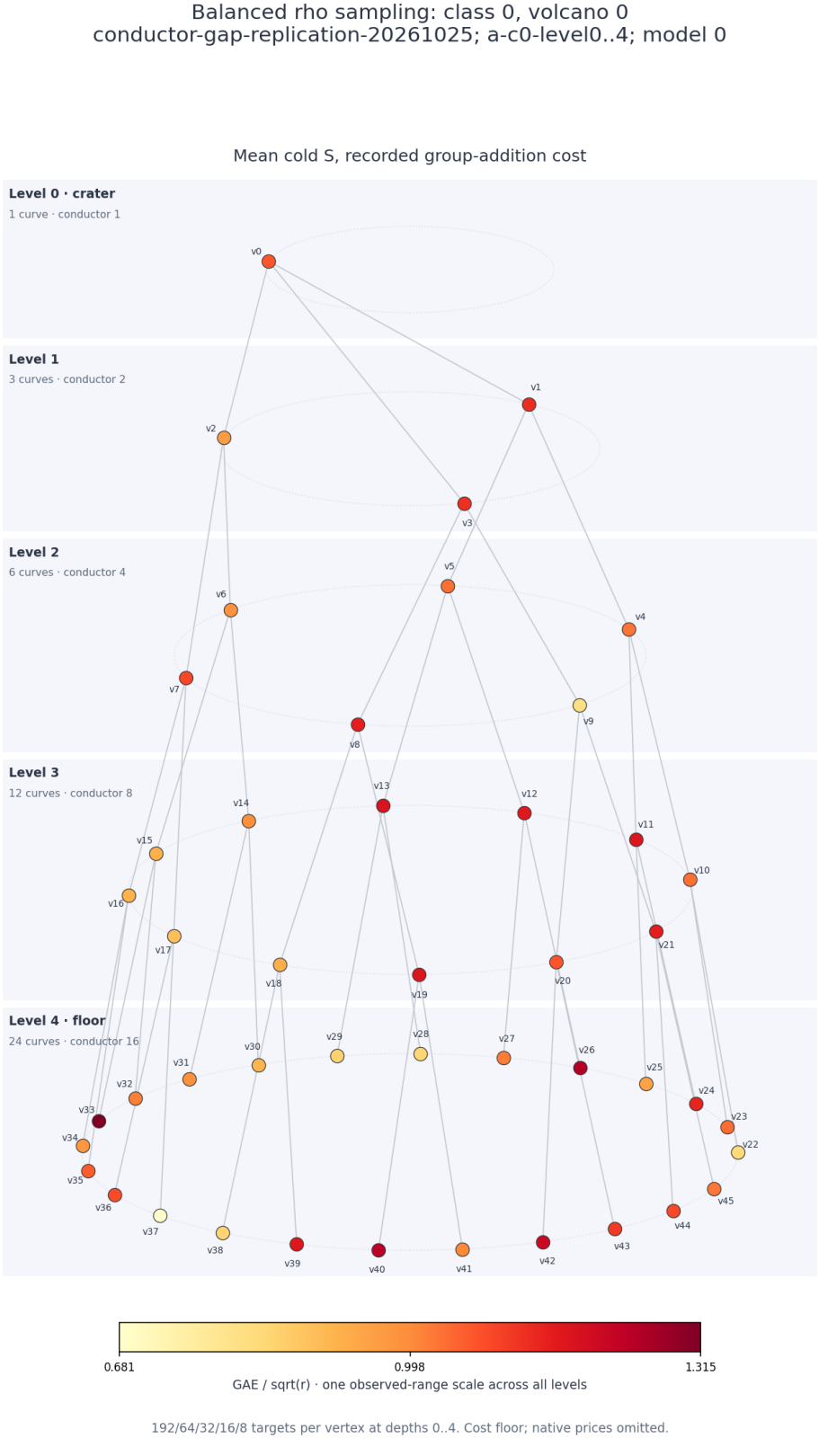


Figure 3: One S1 volcano in A class 0.

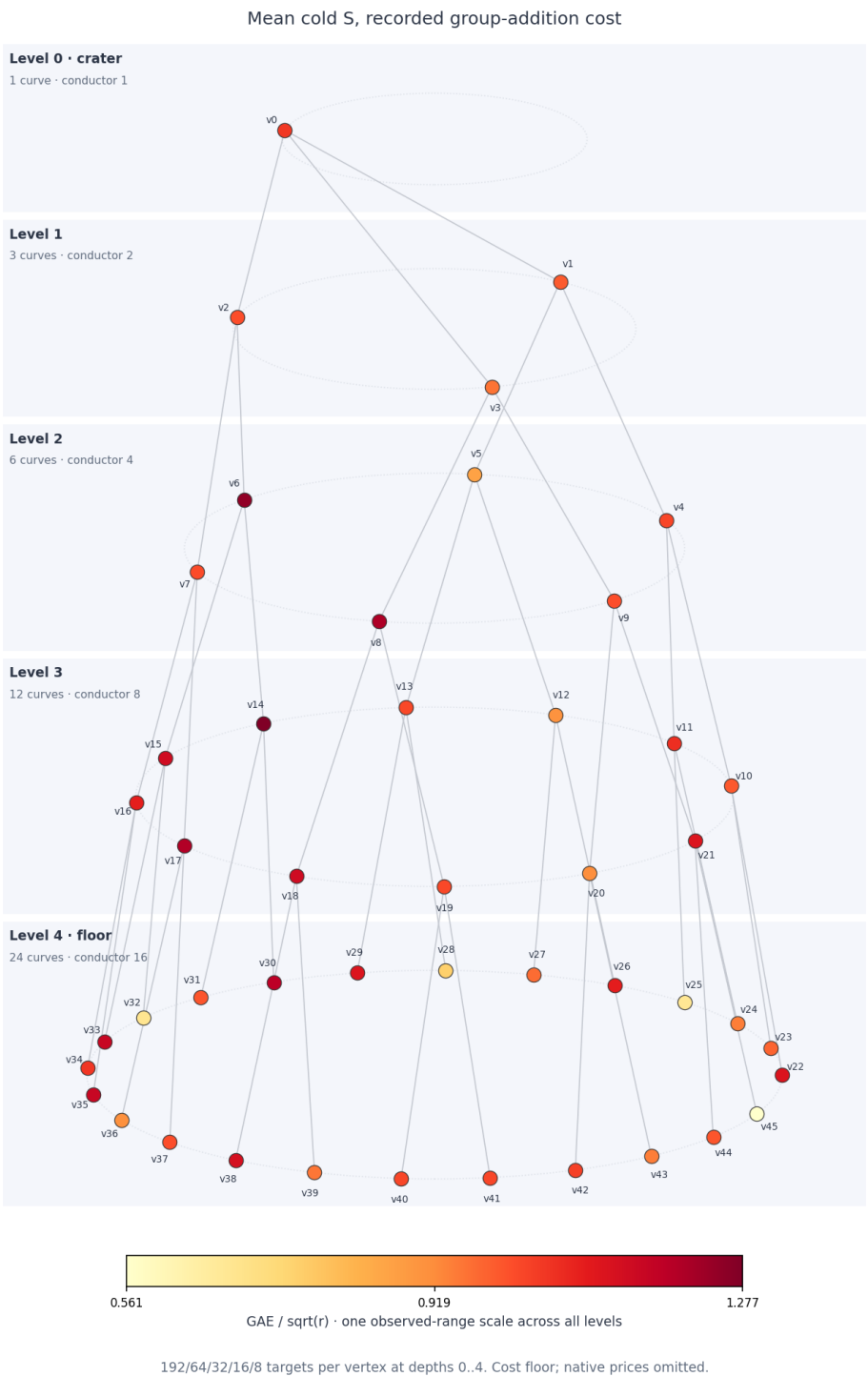


Figure 4: One S2 volcano across the gap in the same class.

receipt's legacy-preservation assertion was insufficient and is explicitly retracted in REGISTRY_CORRECTION.md. The frozen measurements, identities and preflight artifacts remain intact.

All rows use separately executed one-target cold solves with fresh setup; each invocation owns its tables and solver state. Transported scalar indices follow the fixed shared law. Group-operation cost excludes cross-gap parameter discovery, map construction and transport. Their native receipts are in the frozen pair/export files; target scalars are shared to construct corresponding known-answer workloads without timing a map as free. The parent's warm-transfer and exponent claims are corrected explicitly in CLAIM_CORRECTIONS.md.

Raw sessions are preserved byte-for-byte in the deterministic archive `runs.tar.xz` (contains `sessions/`). From this study directory:

```
sha256sum -c runs.tar.xz.sha256
tar -xJf runs.tar.xz
```

From the repository root, rebuild the frozen measurement commit using the recorded compiler; run ecbench verify with –replay-all –exit-code on each extracted session. The audit workflow contains the complete loop. The native example's analyze/diagnostics modes regenerate summaries into fresh output paths; independent.cpp and independent_geometry.cpp supply the separate arithmetic checks. The analysis CSVs carry exact curve, generator, target and recovered scalar values for independent recomputation.

Every session has its own descriptive report, recorded-value figure and inspected PDF under reports/. The raw records and auditor receipts are never edited. Source generators: render.py, validation-inputs.mjs and this metadata-only report renderer. No complete-cost IC/rho leaderboard or benchmark row is promoted without the existing validation gate. The open obligations remain in the conductor-gap problem record.